



# Name S. Sridharshan

B.Tech. in Chemical Engineering

Minor in Green and Renewable Energy (On-going)

Indian Institute of Technology, Tirupati

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## Education

| Degree/Certificate | Institute/Board                          | CGPA/Percentage | Year      |
|--------------------|------------------------------------------|-----------------|-----------|
| B.Tech. (CHE)      | Indian Institute of Technology, Tirupati | 8.44 / 3.376    | 2023-2027 |
| Senior Secondary   | STATE Board                              | 92.5%           | 2022      |
| Secondary          | STATE Board                              | 86.4%           | 2020      |

## Experience

- Research and Development Intern – Avkalan Labs** Dec 2025 – Present
  - Working on physics-informed neural networks (PINNs) for battery modelling, combining electrochemical equations with deep learning to capture cell behaviour under dynamic loading.
  - Building hybrid models that use both physics and data to estimate battery states in real time, with a focus on improving accuracy and robustness over purely data-driven approaches.
- Research Intern – Robbell Materials Research Centre (OPC) Pvt. Ltd.** May 2025 – June 2025
  - Developed a custom cathode material ( $K^+/Cu^{2+}$  co-doped  $Mn_3O_4$ ) for aqueous Zn-ion batteries and evaluated its electrochemical behaviour.
  - Performed cyclic voltammetry (CV) and galvanostatic charge–discharge (GCD) tests.

## Projects

- Physics-Based Thermal Modelling of a 21700 Li-ion Cell using PyBaMM** Oct 2025 – Present
  - Replicating a published COMSOL-based electro-thermal model of a 21700 Li-ion cell entirely in PyBaMM, to study heat generation and temperature rise during cycling.
- Ecology Course project: Development of Personal Water Treatment Systems for Sustainable Water Quality Enhancement within IIT Tirupati Campus** Jan. 2025 - May 2025
  - Proposed and defined the concept for a real-time water-quality monitoring app (TDS and pH), then coordinated with teammates who implemented the Android application.
  - Worked with the team to integrate sensor readings into the app and validate measurements at campus drinking water points.
  - Independently designed and fabricated a rainwater harvesting stand to collect water from dripping umbrellas, aligning the project with sustainability objectives.
- Applied Process Engineering – Linde Ammonia Concept (LAC) Analysis** May 2025
  - Analyzed the Linde Ammonia Concept process step-by-step, connecting unit operations to concepts such as solution thermodynamics, heat integration, and reaction engineering.

## Skills

- Battery & Modelling**
  - Physics-based Li-ion cell modelling (PyBaMM, DFN/SPM, electro-thermal coupling)
  - Battery pack modelling and state estimation (SoC/SoH) in MATLAB/Simulink and Simscape Battery
  - Basic understanding of BMS architecture, safety limits, and control logic
- Programming & Data**
  - Python (NumPy, Matplotlib, basic data handling)
  - MATLAB (Simulink, Simscape Battery, control and simulation workflows)
  - Data analysis and visualization using Excel and Python

## Key Courses Taken

- Battery Manufacturing Technology, Electrochemical Energy Storage, Computational Techniques for Chemical Engineers, Chemical Reaction Engineering, Process Control and Dynamics, Introduction to Renewable Energy.

## Position of Responsibility

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|---------------------------------------------------------------------------------------------------|------------------------------|
| • <b>Co-Coordinator (Event Management)</b> , Chemical Engineering Club, IIT Tirupati              | <i>Jul. 2025 – present</i>   |
| • <b>The Captain</b> , IIT Tirupati Athletics Team, IIT Tirupati                                  | <i>Aug. 2025 – present</i>   |
| • <b>Course Project Group Head</b> , Ecology course Project (a group of 15 members), IIT Tirupati | <i>Jan. 2025 - May. 2025</i> |
| • <b>The Vice Captain</b> , Dragoons Sports team, Kridhan-2025, IIT Tirupati                      | <i>Mar. 2025 - Apr. 2025</i> |

## Certificates

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- **Battery and Battery Management Systems** – DIYGURU, July 2025
  - **Battery Pack Modeling** – MathWorks, July 2025
  - **Battery State Estimation**– MathWorks, July 2025
  - **Simscape Battery**– MathWorks, July 2025

## Achievements

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- **Secured 1st Position:** Poster Presentation (Linde Ammonia Concept) organized by Chemical Engineering Club, IIT Tirupati. The presentation focused on “The Linde Ammonia Concept,” demonstrating a practical understanding of solution thermodynamics, heat transfer, and process calculations. Apr 2025
  - **Secured 2nd Position:** “D’Solve It” - 24-hour Simulink Hackathon organized by Chemical Engineering Club, IIT Tirupati. The challenge involved developing a pH Process Simulator and designing a Feedback Controller to regulate pH. Nov 2025